Where does this vector live?

```
std::vector<int> v = {1, 2, 3};
```

STACK or **HEAP**

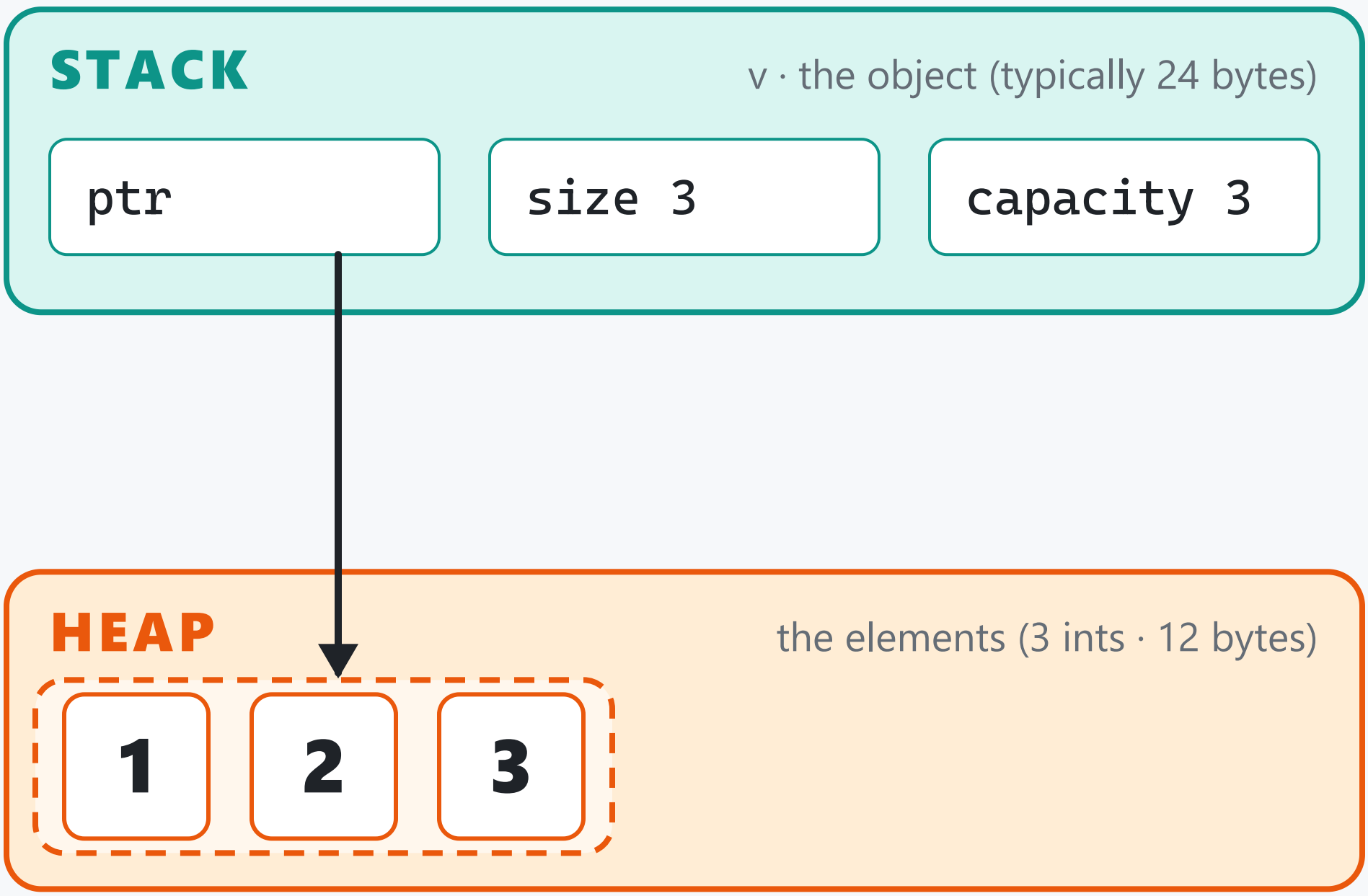
Pick one.

There's a catch.

swipe →

What's actually in memory

```
std::vector<int> v = {1, 2, 3};
```

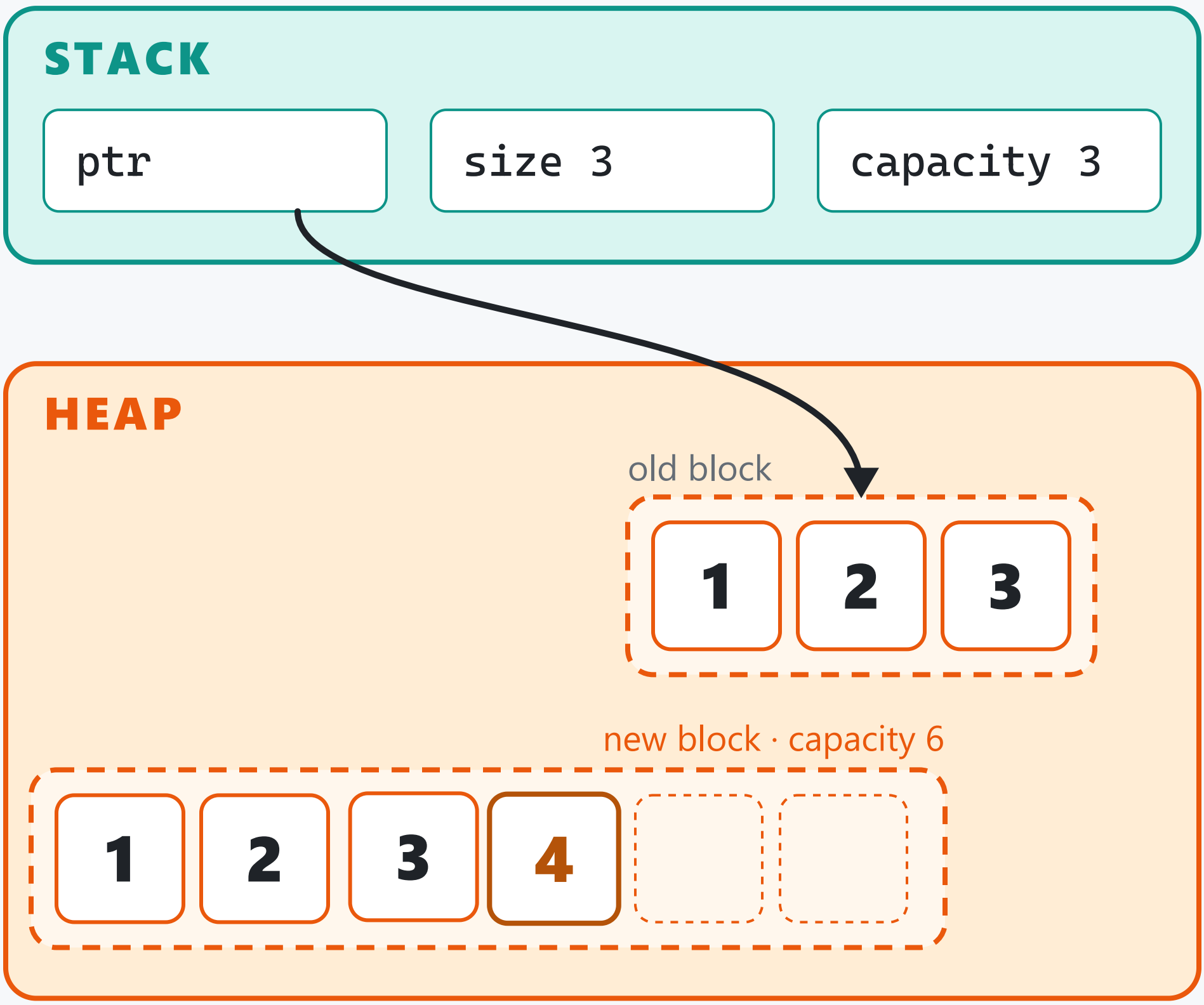


The object → stack
Its elements → heap

So when it grows,
it has to move.

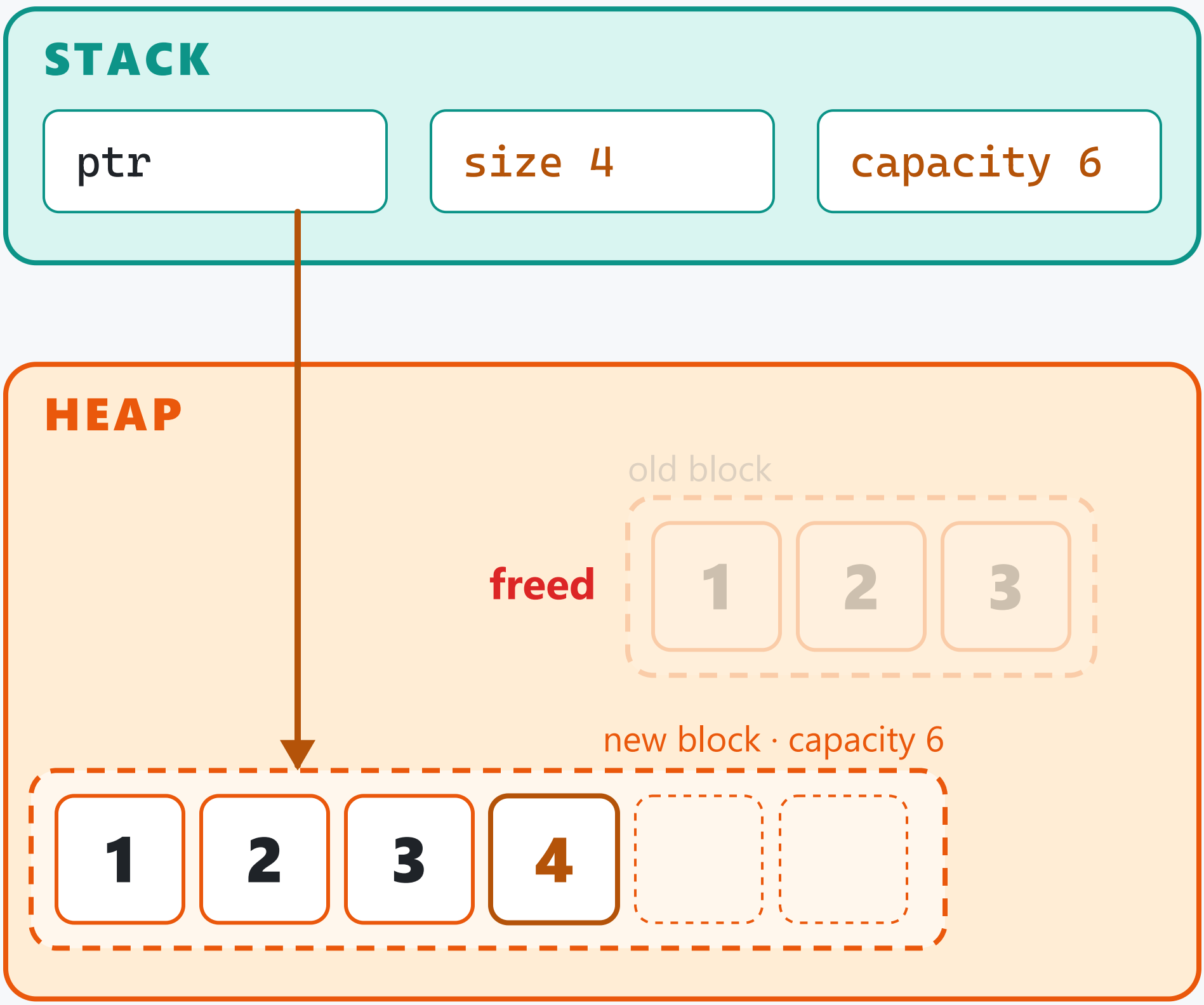
```
v.push_back(4);
```

capacity is full



```
v.push_back(4);
```

capacity is full

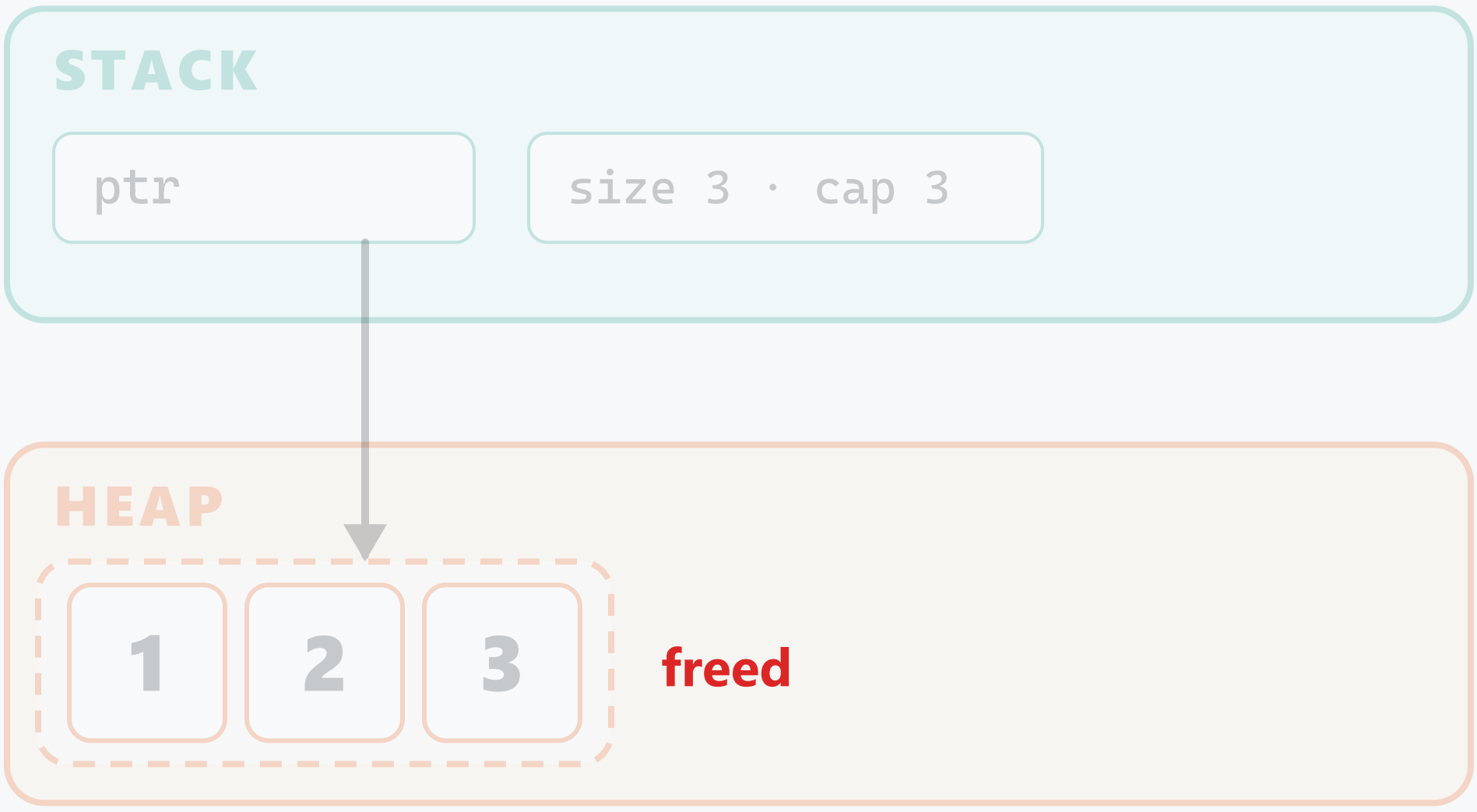


The old block is gone.

Every iterator and pointer into it now dangles.

Growth factor depends on the implementation.

```
{
std::vector<int> v = {1, 2, 3};
} // v is destroyed here
```



`~vector()` runs

it frees the heap block for you

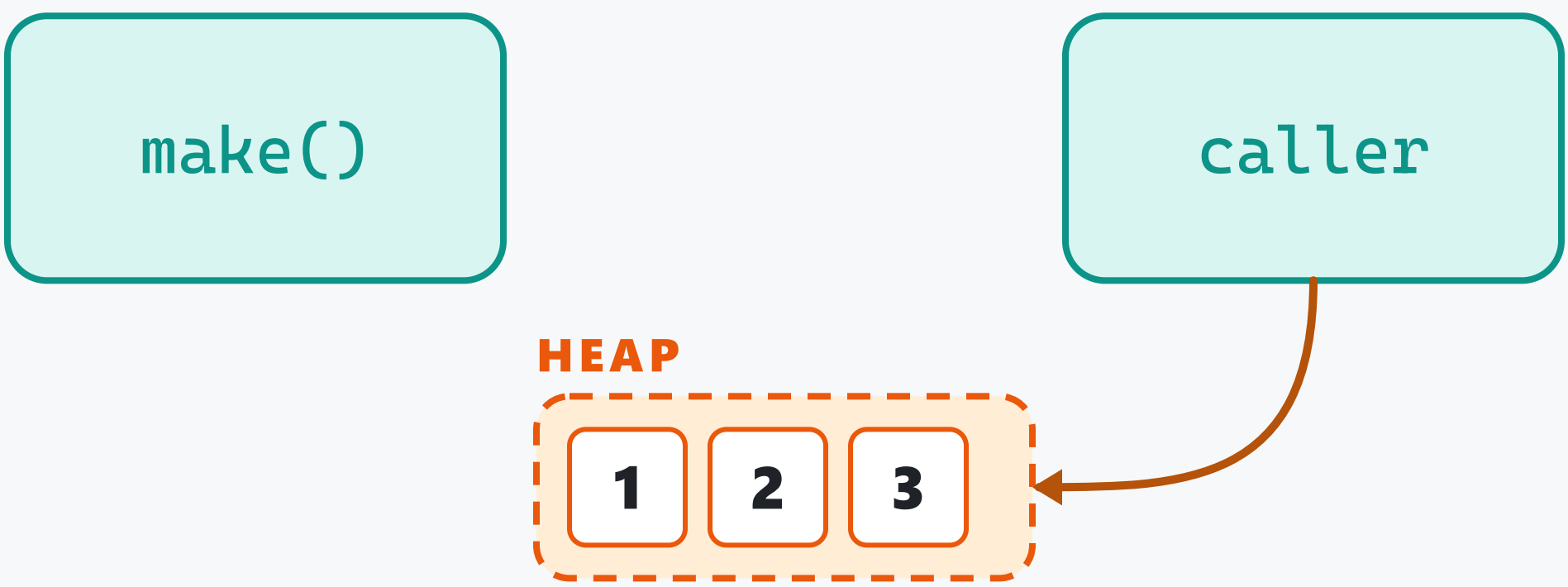
No delete. No leak.

That's RAII.

The rule

Return it by value.

```
std::vector<int> make();
```



same heap block, new owner · no deep copy

By value or by pointer?

What do you do?